Fiscal Year:	FY 2019	**Task Last Updated:**	FY 10/19/2020
PI Name:	Urbaniak, Camilla Ph.D.		
Project Title:	Characterization of the Environmental Microbiome of the International Space Station		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	camilla.urbaniak@jpl.nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	626-840-6992
Organization Name:	ZIN Technologies, Inc		
PI Address 1:	4800 Oak Grove Dr		
PI Address 2:			
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:	NOTE: PI had NASA Postdoctoral Program Fellowship at NASA Jet Propulsion Laboratory until 2019; now with Zin Technologies, Inc., Middleburg, OH, but remains physically at NASA JPL address.		
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	08/01/2016	**End Date:**	07/31/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Administration, USRA	**Contact Phone:**	
Contact Email:	npphelp@usra.edu		
Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2019 per JPL (Ed., 12/31/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Venkateswaran, Kasthuri Ph.D. (MENTOR/ NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Postdoctoral Program Fellow Using next generation sequencing and microarrays, the microbial communities present on the International Space Station (ISS), over a period of one year, will be characterized. Factors affecting this microbial composition and those that promote pathogen transmission between surfaces and the crew will be assessed. In addition, the effects of spaceflight on antibiotic resistance and virulence factors will be examined. Findings from this study will allow mission planners to design better strategies to mitigate health risk to astronauts while in Space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	ED. NOTE October 2020: Final report not received from USRA; added articles acknowledging this project in Bibliography section.
Bibliography Type:	Description: (Last Updated: 11/01/2024)
Articles in Peer-reviewed Journals	Checinska Sielaff A, Urbaniak C, Mohan GBM, Stepanov VG, Tran Q, Wood JM, Minich J, McDonald D, Mayer T, Knight R, Karouia F, Fox GE, Venkateswaran K. "Characterization of the total and viable bacterial and fungal communities associated with the International Space Station surfaces." Microbiome. 2019 Apr 8;7(1):50. https://doi.org/10.1186/s40168-019-0666-x ; PubMed PMID: 30955503 ; PubMed Central PMCID: PMC6452512 , Apr-2019
Articles in Peer-reviewed Journals	Urbaniak C, van Dam P, Zaborin A, Zaborina O, Gilbert JA, Torok T, Wang CCC, Venkateswaran K "Genomic characterization and virulence potential of two Fusarium oxysporum isolates cultured from the International Space Station." mSystems. 2019 Mar 19;4(2):e00345-18. https://doi.org/10.1128/mSystems.00345-18 ; PubMed PMID: 30944876 ; PubMed Central PMCID: PMC6426649 , Jan-2020
Articles in Peer-reviewed Journals	Malli Mohan GB, Parker CW, Urbaniak C, Singh NK, Hood A, Minich JJ, Knight R, Rucker M, Venkateswaran K. "Microbiome and metagenome analyses of a closed habitat during human occupation." mSystems. 2020 Jul 28;5(4):e00367-20. https://doi.org/10.1128/mSystems.00367-20 ; PMID: 32723791; PMCID: PMC7394354 , Jul-2020
Articles in Peer-reviewed Journals	Urbaniak C, Lorenzi H, Thissen J, Jaing C, Crucian B, Sams C, Pierson D, Venkateswaran K, Mehta S. "The influence of spaceflight on the astronaut salivary microbiome and the search for a microbiome biomarker for viral reactivation." Microbiome. 2020 Apr 20;8(1):56. https://doi.org/10.1186/s40168-020-00830-z ; PMID: 32312311; PMCID: PMC7171750 , Apr-2020
Articles in Peer-reviewed Journals	Urbaniak C, Wong S, Tighe S, Arumugam A, Liu B, Parker CW, Wood JM, Singh NK, Skorupa DJ, Peyton BM, Jenson R, Karouia F, Dragon J, Venkateswaran K. "Validating an automated nucleic acid extraction device for Omics in Space using whole cell microbial reference standards." Front Microbiol. 2020 Aug 21;11:1909. https://doi.org/10.3389/fmicb.2020.01909 ; PMID: 32973700; PMCID: PMC7472602 , Aug-2020
Articles in Peer-reviewed Journals	Morrison MD, Thissen JB, Karouia F, Mehta S, Urbaniak C, Venkateswaran K, Smith DJ, Jaing C. "Investigation of spaceflight induced changes to astronaut microbiomes." Front Microbiol. 2021 Jun 2;12:659179. https://doi.org/10.3389/fmicb.2021.659179 ; PMID: 34149649; PMCID: PMC8207296 , Jun-2021
Articles in Peer-reviewed Journals	Simpson AC, Urbaniak C, Bateh JR, Singh NK, Wood JM, Debieu M, O'Hara NB, Houbraken J, Mason CE, Venkateswaran K. "Draft genome sequences of fungi isolated from the International Space Station during the Microbial Tracking-2 experiment." Microbiol Resour Announc. 2021 Sep 16;10(37):e0075121. https://doi.org/10.1128/MRA.00751-21 ; PMID: 34528817; PMCID: PMC8444978 , Sep-2021
Articles in Peer-reviewed Journals	Urbaniak C, Morrison MD, Thissen JB, Karouia F, Smith DJ, Mehta S, Jaing C, Venkateswaran K. "Microbial tracking-2, a metagenomics analysis of bacteria and fungi onboard the International Space Station." Microbiome. 2022 Jun 29;10(1):100. https://doi.org/10.1186/s40168-022-01293-0 ; PMID: 35765106; PMCID: PMC9241228 , Jun-2022
Articles in Peer-reviewed Journals	Wood JM, Urbaniak C, Parker C, Singh NK, Wong S, Arumugam A, Skorupa DJ, Hemmah A, Laaguiby P, Karouia F, Peyton BM, Tighe S, Venkateswaran K. "Assessing microbial diversity in Yellowstone National Park hot springs using a field deployable automated nucleic acid extraction system." Front Ecol Evol. 2024 Feb 9;12:1306008. https://doi.org/10.3389/fevo.2024.1306008 , Feb-2024